

Somaiya Vidyavihar University
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Batch: A-4 **Roll No.:** 16010122151

Experiment / Assignment / Tutorial No: 6

Title: Random Number Generation and Hypothesis Testing

Objective: The objective of this lab experiment is to generate random arrival times for vehicles using a Linear Congruential Generator (LCG), simulate vehicle arrivals at an intersection with traffic light control, and perform hypothesis testing on the generated random numbers to ensure their randomness and uniformity using Python SimPy discrete-event simulation library.

Expected Outcome of Experiment:

CO3: Generate pseudorandom numbers and perform statistical tests to measure the quality of a pseudorandom number generator.

Books/ Journals/ Websites referred:

1. “Discrete-Event System Simulation” by Jerry Banks, John S. Carson II, Barry L. Nelson, David M. Nicol.
2. SimPy Documentation: <https://simpy.readthedocs.io/en/latest/>
3. SciPy Documentation: <https://docs.scipy.org/doc/scipy/>

Background:

(Explain in brief Random number generation techniques and Chi-square, ks and runs up and down Hypothesis testing)

Problem Statement:

1. Random Number Generation for Vehicle Arrivals:

- Implement a Linear Congruential Generator (LCG) to generate uniform random numbers.
- Transform these uniform random numbers into exponential inter-arrival times to model a Poisson arrival process.

2. Visualization:

- Plot a histogram of the generated exponential inter-arrival times to visualize their distribution.

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3. Simulation of Traffic Flow:

- Using the generated arrival times, simulate the traffic flow at an intersection controlled by a traffic light.
- Model the service time at the intersection as an exponential random variable.
- Control the traffic light with alternating green and red phases.

4. Testing Random Numbers:

- Perform a Chi-square test to compare the observed frequency distribution of the uniform random numbers to the expected uniform distribution.
- Conduct a Kolmogorov-Smirnov (K-S) test to compare the distribution of the generated random numbers to a uniform distribution.
- Execute a Runs up and down test to check the randomness of the sequence by counting the number of runs (increasing or decreasing sequences).

5. Analysis

- Analyze the results of the Chi-square test, K-S test, and Runs up and down test to determine the uniformity and randomness of the generated random numbers.
- Discuss the effectiveness of the random number generation and the validity of the traffic simulation model based on these results.

Implementation Steps with Screen shots:

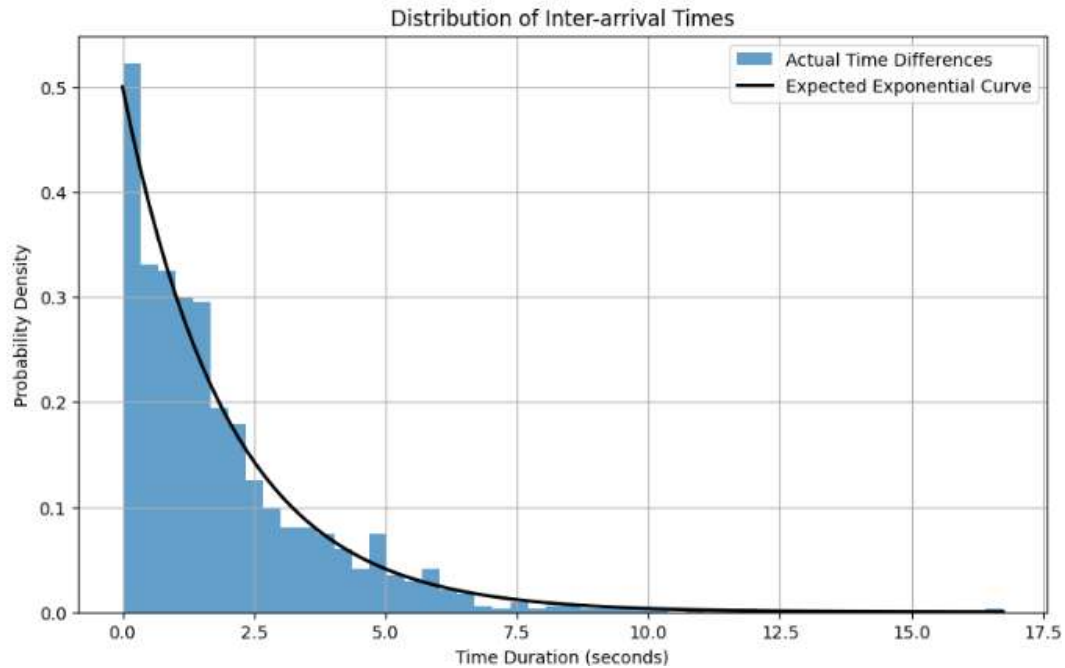
Notebook link : [Expt6.ipynb](#).

OUTPUT:

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Starting Generator Setup
Total uniform numbers produced: 1000.
Initial five uniform values: ['0.2143', '0.8758', '0.5243', '0.3436', '0.5449']

Total exponential arrival times calculated: 1000.
Initial five arrival times (s): ['0.4823', '4.1721', '1.4861', '0.8418', '1.5746']



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Running Traffic Scenario

Time 0.48: Car-1 reaches the crossing.
Time 0.48: Car-1 proceeds on GREEN.
Time 2.15: Car-1 clears the junction. Time taken: 1.67s.
Time 4.65: Car-2 reaches the crossing.
Time 4.65: Car-2 proceeds on GREEN.
Time 5.46: Car-2 clears the junction. Time taken: 0.81s.
Time 6.14: Car-3 reaches the crossing.
Time 6.14: Car-3 proceeds on GREEN.
Time 6.60: Car-3 clears the junction. Time taken: 0.46s.
Time 6.98: Car-4 reaches the crossing.
Time 6.98: Car-4 proceeds on GREEN.
Time 8.07: Car-4 clears the junction. Time taken: 1.09s.
Time 8.56: Car-5 reaches the crossing.
Time 8.56: Car-5 proceeds on GREEN.
Time 9.49: Car-6 reaches the crossing.
Time 9.49: Car-6 proceeds on GREEN.
Time 10.17: Car-7 reaches the crossing.
Time 10.17: Car-7 proceeds on GREEN.
Time 11.15: Car-5 clears the junction. Time taken: 2.59s.
Time 11.17: Car-6 clears the junction. Time taken: 1.68s.
Time 11.70: Car-8 reaches the crossing.
Time 11.70: Car-8 proceeds on GREEN.
Time 11.80: Car-7 clears the junction. Time taken: 1.63s.
Time 12.71: Car-9 reaches the crossing.
Time 12.71: Car-9 proceeds on GREEN.
Time 12.98: Car-10 reaches the crossing.
Time 12.98: Car-10 proceeds on GREEN.
Time 13.57: Car-8 clears the junction. Time taken: 1.87s.
Time 14.10: Car-11 reaches the crossing.
Time 14.10: Car-11 proceeds on GREEN.
Time 14.77: Car-12 reaches the crossing.
Time 14.77: Car-12 proceeds on GREEN.
Time 14.86: Car-9 clears the junction. Time taken: 2.15s.
Time 15.57: Car-13 reaches the crossing.

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Time 494.52: Car-292 stops for the RED light.
Time 495.02: Car-283 proceeds on GREEN.
Time 495.09: Car-288 proceeds on GREEN.
Time 495.15: Car-271 proceeds on GREEN.
Time 495.17: Car-291 proceeds on GREEN.
Time 495.18: Car-290 proceeds on GREEN.
Time 495.21: Car-282 proceeds on GREEN.
Time 495.22: Car-274 proceeds on GREEN.
Time 495.25: Car-289 proceeds on GREEN.
Time 495.30: Car-272 proceeds on GREEN.
Time 495.35: Car-275 proceeds on GREEN.
Time 495.44: Car-284 proceeds on GREEN.
Time 495.52: Car-292 proceeds on GREEN.
Time 495.56: Car-270 clears the junction. Time taken: 27.73s.
Time 495.57: Car-279 proceeds on GREEN.
Time 495.58: Car-277 proceeds on GREEN.
Time 495.62: Car-287 proceeds on GREEN.
Time 495.65: Car-280 proceeds on GREEN.
Time 495.66: Car-281 proceeds on GREEN.
Time 495.83: Car-286 proceeds on GREEN.
Time 495.83: Car-283 clears the junction. Time taken: 12.81s.
Time 495.84: Car-278 proceeds on GREEN.
Time 495.85: Car-285 proceeds on GREEN.
Time 495.87: Car-273 proceeds on GREEN.
Time 496.00: Car-276 proceeds on GREEN.
Time 496.25: Car-288 clears the junction. Time taken: 7.16s.
Time 496.25: Car-271 clears the junction. Time taken: 25.11s.
Time 496.57: Car-291 clears the junction. Time taken: 4.40s.
Time 499.44: Car-293 reaches the crossing.
Time 499.44: Car-293 proceeds on GREEN.
Time 499.53: Car-290 clears the junction. Time taken: 8.34s.
```

Section 4: Statistical Checks

Chi-Square Score: 12.3800

P-Value from Chi-Square: 0.1927

Decision (0.05 level): We find no evidence to reject uniformity.

Kolmogorov-Smirnov Score: 0.0309

P-Value from K-S: 0.2878

Decision (0.05 level): Data appears consistent with uniform distribution.

Section 5: Randomness Check

Total Runs Counted: 648

Z-Score for Runs: -1.3762

P-Value for Runs: 0.1687

Decision (0.05 level): The sequence seems random.

Conclusion: Thus, from this experiment, we learnt about Random Number Generation and Hypothesis Testing, implementing a running traffic scenario while plotting a distribution of inter-arrival times & after a randomness check, coming to a conclusion.

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Post lab Questions:

1. Explain Gap & Poker test with the help of example.

Gap Test

The **Gap Test** checks for the randomness of digits by analyzing the "gaps" or distances between consecutive occurrences of a specific digit. If the numbers are random, the lengths of these gaps should follow a predictable geometric distribution.

How It Works

1. **Choose a Digit:** Select a digit (e.g., from 0-9) to analyze.
2. **Record Gap Lengths:** Go through your sequence of numbers and record the length of the "gaps" between every appearance of your chosen digit. A gap of length 0 means the digit appeared twice in a row (e.g., ... 4, 4 ...).
3. **Count Frequencies:** Tally how many times each gap length occurred. This gives you the **Observed Frequencies** (O_i).
4. **Calculate Expected Frequencies:** For a truly random sequence, the probability of a gap of length x is given by the formula $P(x) = (0.9)^x(0.1)$. The **Expected Frequency** (E_i) for N total gaps is $E_i = N \cdot P(x)$.
5. **Apply Chi-Square Test:** Use the Chi-Square formula to compare observed vs. expected values:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

If the calculated χ^2 value is less than a critical value from a Chi-Square table, we cannot reject the hypothesis that the sequence is random.

Example

Let's test the randomness of the digit '7' in the following sequence of 50 numbers.

Sequence: 4, 9, 1, 7, 2, 5, 3, 7, 7, 0, 8, 1, 7, 2, 6, 6, 9, 7, 5, 1, 3, 4, 7, 8, 8, 2, 0, 1, 7, 3, 5, 2, 9, 7, 6, 7, 4, 1, 9, 0, 7, 5, 5, 8, 6, 7, 3, 3, 1, 0

1. **Identify the Gaps for digit '7':**
 - 7, 2, 5, 3, 7 → Gap length is 3.
 - 7, 7 → Gap length is 0.
 - 7, 0, 8, 1, 7 → Gap length is 3.

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- 7, 2, 6, 6, 9, 7 → Gap length is 4.
- 7, 5, 1, 3, 4, 7 → Gap length is 4.
- 7, 8, 8, 2, 0, 1, 7 → Gap length is 5.
- 7, 3, 5, 2, 9, 7 → Gap length is 4.
- 7, 6, 7 → Gap length is 1.
- 7, 4, 1, 9, 0, 7 → Gap length is 4.
- 7, 5, 5, 8, 6, 7 → Gap length is 4.

We found a total of **N = 10** gaps.

2. Create Frequency Table:

Gap Length (x)	Observed Frequency (O _i)	Probability P(x)=(0.9)x(0.1)	Expected Frequency (E _i =10·P(x))	E _i (O _i -E _i) ²
0	1	0.100	1.00	0.000
1	1	0.090	0.90	0.011
2	0	0.081	0.81	0.810
3	2	0.073	0.73	2.149
4	5	0.066	0.66	28.327
5	1	0.059	0.59	0.288
≥6	0	0.531	5.31	5.310
Total	10	-	10.00	χ²=36.895

3. **Conclusion:** Let's assume a significance level $\alpha=0.05$. For 6 degrees of freedom (7 categories - 1), the critical value from a χ^2 table is **12.59**.

Since our calculated value $\chi^2=36.895$ is **much greater than** the critical value of 12.59, we **reject the hypothesis** that the numbers are random with respect to the distribution of the digit '7'. The high contribution from "Gap Length 4" suggests a non-random pattern.

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Poker Test

The **Poker Test** checks for randomness by grouping numbers into "hands" (like in poker) and comparing the frequencies of hand types (e.g., one pair, three of a kind, all different) to what's expected in a random sequence. It's a good test for independence between numbers.

How It Works

1. **Group Numbers:** Divide the sequence into blocks of a fixed size (e.g., 5 digits per block).
2. **Categorize Hands:** Classify each block into a poker-hand category. For 5-digit hands with digits 0-9, common categories are:
 - **All Different:** (e.g., 12345)
 - **One Pair:** (e.g., 11234)
 - **Two Pairs:** (e.g., 11223)
 - **Three of a Kind:** (e.g., 11123)
 - **Full House:** (e.g., 11122)
 - **Four of a Kind:** (e.g., 11112)
 - **Five of a Kind:** (e.g., 11111)
3. **Count Frequencies:** Tally the number of hands in each category to get the **Observed Frequencies (O_i)**.
4. **Calculate Expected Frequencies:** Use combinatorics to find the theoretical probability of each hand type. The **Expected Frequency (E_i)** is the total number of hands (N) multiplied by that probability.
5. **Apply Chi-Square Test:** Use the same χ^2 formula as before to compare observed and expected values.

Example

Let's test a sequence of **1000 numbers**, which we'll group into $N = 200$ hands of 5 digits each.

1. **Observed Frequencies (Hypothetical):** After analyzing our 200 hands, we get the following counts:

Hand Category	Observed Frequency (O_i)
All Different	65

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Hand Category	Observed Frequency (O_i)
One Pair	98
Two Pairs	23
Three of a Kind	12
Full House	1
Four/Five of a Kind	1
Total	200

Note: We've combined the rarest categories (Four of a Kind, Five of a Kind) because the Chi-Square test works best when expected frequencies are not too small (typically > 5).

2. **Calculate Expected Frequencies:** Here are the standard probabilities for 5-digit hands and the resulting expected frequencies for $N=200$ hands.

Hand Category	Probability (p_i)	Expected Frequency ($E_i = 10 \cdot P(x)$)	$\frac{(O_i - E_i)^2}{E_i}$
All Different	0.3024	60.48	0.337
One Pair	0.5040	100.80	0.078
Two Pairs	0.1080	21.60	0.091
Three of a Kind	0.0720	14.40	0.400
Full House	0.0090	1.80	0.356
Four/Five of a Kind	0.0046	0.92	0.007
Total	1.0000	200.00	$\chi^2=1.269$

3. **Conclusion:** Let's use a significance level $\alpha=0.05$. For 5 degrees of freedom (6 categories - 1), the critical value from a χ^2 table is **11.07**.

Since our calculated value $\chi^2=1.269$ is **less than** the critical value of 11.07, we **do not reject the hypothesis** that the sequence is random. The observed frequencies are close enough to what we would expect from a truly random sequence.

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2. Consider the multiplicative Congruential generator under the following circumstances:

- a) $X_0 = 7, a = 11, m = 16$
- b) $X_0 = 8, a = 11, m = 16$
- c) $X_0 = 7, a = 7, m = 16$
- d) $X_0 = 8, a = 7, m = 16$

Generate enough values in each case to complete a cycle. What inferences can be drawn? Is maximum period achieved?

CODE : Attached

OUTPUT:



```
Analyzing MCG with X0=7, a=11, m=16
Generated Sequence: [7, 13, 15, 5]
Cycle repeats with value: 7
Period Length: 4
Max Possible Period for m=16: 4.0
Result: Maximum period was achieved.

Analyzing MCG with X0=8, a=11, m=16
Generated Sequence: [8]
Cycle repeats with value: 8
Period Length: 1
Max Possible Period for m=16: 4.0
Result: Maximum period was NOT achieved.

Analyzing MCG with X0=7, a=7, m=16
Generated Sequence: [7, 1]
Cycle repeats with value: 7
Period Length: 2
Max Possible Period for m=16: 4.0
Result: Maximum period was NOT achieved.

Analyzing MCG with X0=8, a=7, m=16
Generated Sequence: [8]
Cycle repeats with value: 8
Period Length: 1
Max Possible Period for m=16: 4.0
Result: Maximum period was NOT achieved.
```